O17e Diffraction

Tasks

0. During the preparation for this experiment, analytically calculate the Fourier transform of a single-slit and a double-slit function and compare these with the familiar Fraunhofer diffraction-patterns for these objects. Using FFT (Fast-Fourier-Transform), determine the Fourier transforms of the functions $F(k) = \mathrm{sinc}(2\pi k) = \sin(2\pi k)/(2\pi k)$ and $F(k)^2$. Discuss this result in the light of the convolution theorem.

1. Measure the diffraction pattern of a single slit in the Fraunhofer regime.
Determine the slit width b
(a) from the position of the minima for various diffraction orders
(b) from the FFT of the intensity profile.

2. Measure the diffraction pattern of a double slit in the Fraunhofer regime.
Determine
(a) the slit distance g from the position of the minima for various diffraction orders
(b) the slit distance g and the slit width b from the fit of the Fraunhofer diffraction-pattern to the data
(c) the slit distance g and the slit width b from the FFT of the intensity profile.

3. Determine the grating constant and the Talbot length of a diffraction grating. Take diffraction images of the grating at various distances between the camera and the grating in the Fresnel regime. Calculate the FFT and determine the intensity of the first-order peak as a function of distance. Use the intensity oscillations to determine the Talbot length. Compare the experimental and theoretical values.

Accessories

Diode laser (λ = 636 nm), optical bench, CCD line camera (Thorlabs), polarizer, lens, single and double slit, diffraction grating, PC

Keywords for preparation

- Interference, coherence, coherence condition
- Diffraction at single slit and double slit
- Fraunhofer and Fresnel diffraction
- Fourier transformation

Literature

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Physikalisches Praktikum, 14. Auflage, Hrsg. W. Schenk, F. Kremer, Optik, 2.0, 2.2, 2.3, Fourier-Transformation und Signalanalyse, 1.0, https://link.springer.com/book/10.1007/978-3-658-00666-2

Safety Note

The primary beam of the laser (class 2) has a power output of approximately 1 mW. Looking directly into the beam or its specular reflection may cause eye injury. The beam is not shielded. During experimentation, do not look directly into the beam or a reflection from a shiny surface.

For the observation of the interference patterns with a CCD camera, one or two (depending on the type of camera) polarizers have to be placed between laser and slit to attenuate the light (Malus' law).

Fraunhofer and Fresnel diffraction

In this experiment, the diffraction patterns generated by single and double slits are measured and analyzed. According to the Huygens principle, every point of a wave front emits a spherical wave

$$\psi_s = A\frac{\exp(ikr - i\omega t)}{r} \tag{1}$$

such that the wave front at a later time is formed by superposition of all these spherical waves. In Eq. (1), r denotes the distance between emission and observation points, t the time, $k = 2\pi/\lambda$ the modulus of the wave vector, ω the angular frequency, and A the amplitude. ψ stands for a component of the electric or magnetic field.

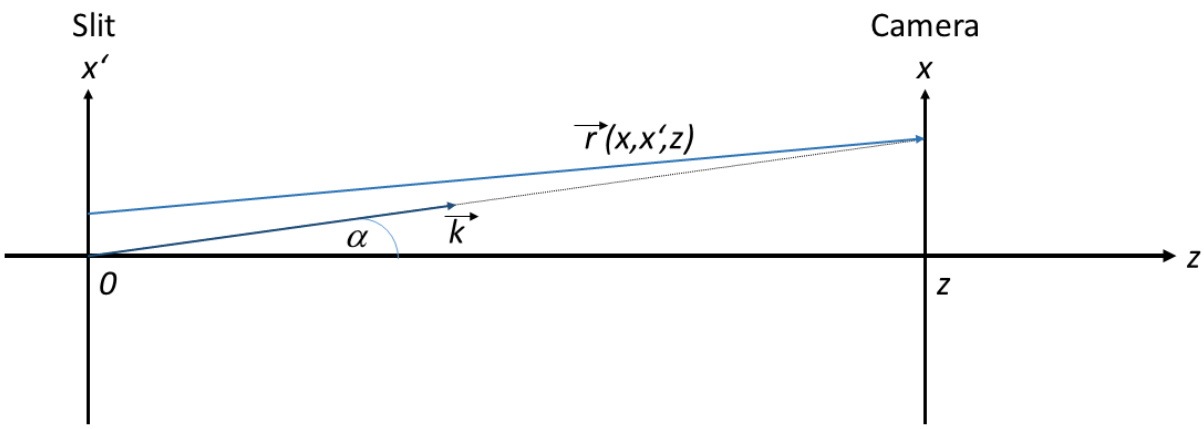

Fig. 1. Optical bench (*z*-axis), coordinates *x'* (slit) and *x* (camera), wave vector $\vec{k}$ with deflection angle α.

In the case of diffraction, this principle might be directly applied as follows, see Fig. 1. On an optical bench a slit/double slit is placed which is illuminated with laser light coming from the left. The diffraction pattern is recorded by a camera to the right of the slit/double slit. A coordinate system is chosen in such a way that the *z*-axis is parallel to the optical bench with the slit/double slit positioned at $z = 0$ and the camera positioned at *z*. The *x'*-*y'* coordinate plane lies in the plane of the slit/double slit perpendicular to the optical bench, such that the transmission function $f(x')$ of the single slit is given by

$f(x') = 1$: $-b/2 < x' < b/2$, $f(x') = 0$ elsewise,

and of the double slit by

$f(x') = 1$: $-(g+b)/2 < x' < -(g-b)/2$ and $(g-b)/2 < x' < (g+b)/2$, $f(x') = 0$ elsewise.

b denotes the slit width and g the slit distance.

The x-y- coordinate plane is in the plane of the camera. If the amplitudes of the spherical waves in the slit/double slit plane are given by $A(x',y')$, then the wave arriving on the camera at the point (x,y) might be written as

$$\psi(x,y,t) = \int A(x',y') \frac{\exp(ikr - i\omega t)}{r} dx'dy' \tag{2}$$

with

$$r = \sqrt{(x-x')^2 + (y-y')^2 + z^2} \ . \tag{3}$$

Equation (2) is simpler than Kirchhoff's diffraction integral, but is sufficient for this discussion. In the following, it is assumed that the slits are infinitely extended along the y'-axis, such that $A(x',y') = A(x')$ and the y'-coordinate can be integrated out.

The integral in Eq. (2) looks deceptively simple, but is difficult to handle because of the square root in r, see Eq. (3). Therefore, one often considers limiting cases, in which Eq. (2) is approximated and can be more conveniently integrated. The limiting cases discussed here are the *Fresnel* regime, or far field regime, and the *Fraunhofer* regime, or extreme far field regime.

Let us first turn to the *Fraunhofer* regime. In this case, the camera is so far away from the slit that the spherical waves arriving on the optical sensor can be considered to be plane waves

$$\psi_p = A_p \exp(i\vec{k} \cdot \vec{r} - i\omega t) \ . \tag{4}$$

If the deflection angle is denoted by α, see Fig. 1, then the wave vector $\vec{k}$ is given by

$$\vec{k} = (k\sin\alpha, 0, k\cos\alpha) \tag{5}$$

and with the vector $\vec{r} = (x - x', 0, z)$ connecting the point $(x',0,0)$ in the slit plane with the observation point $(x,0,z)$, Eq. (2) can be replaced by

$$\psi(x,t) = \int A_p(x') \exp(i\vec{k} \cdot \vec{r} - i\omega t)\, dx' = \exp(ik\sin\alpha x + ik\cos\alpha z - i\omega t) \int A_p(x') \exp(-ik\sin\alpha x')\, dx' \ . \tag{6}$$

The latter integral is a Fourier transform, see below, i.e. the diffraction pattern is given by the Fourier transform of the amplitude function $A_p(x')$. With $A_p(x') \propto f(x')$ and the transmission functions given above, the integral can be readily solved to yield the light intensity I on the camera as

$$I \propto \psi\psi^* \propto \mathrm{sinc}^2\left(k_x \frac{b}{2}\right) = \mathrm{sinc}^2\left(\frac{\pi b \sin\alpha}{\lambda}\right) \tag{7}$$

in case of a single slit and

$$I \propto \psi\psi^* \propto \mathrm{sinc}^2\left(k_x \frac{b}{2}\right)\cos^2\left(k_x \frac{g}{2}\right) = \mathrm{sinc}^2\left(\frac{\pi b \sin\alpha}{\lambda}\right)\cos^2\left(\frac{\pi g \sin\alpha}{\lambda}\right) \tag{8}$$

in case of a double slit.

The wave vector component $k_x = k\sin\alpha$ perpendicular to the optical bench is the conjugate variable to the position perpendicular to the optical bench. The intensity I measured by the camera can be regarded as a function of k_x, such that the Fourier transform back into the real space x' would yield the convolution of the slit/double slit transmission function *f*(*x'*) with itself.

In *Fresnel* diffraction, also the far field limit is treated, but the curvature of the wave fronts is retained in a certain approximation. The distance *z* between camera and diffracting object is considered to be large compared to the vertical dimensions *x'*, *y'*, *x*, *y*, such that the square root in Eq. (3) can be expanded to yield

$$r \approx z + \frac{(x-x')^2 + (y-y')^2}{2z} . \tag{9}$$

Inserting this result in Eq. (2), substituting $A(x',y')/r = f(x')/(i\lambda z)$ and integrating over *y'* yields

$$\psi(x,t) = \frac{\exp(ikz - i\omega t)}{\sqrt{i\lambda z}} \int f(x') \exp\left[i\frac{k}{2z}\left(x-x'\right)^2 \right] dx' . \tag{10}$$

This is called a Fresnel transform. Since the transmission functions of slit and double slit are piecewise constant, the integral in Eq. (10) can be related to the Fresnel integrals

$$F(u) = \int_0^u \exp\left(i\pi x^2/2\right) dx = C(u) + iS(u) = \int_0^u \cos\left(i\pi x^2/2\right) dx + i\int_0^u \sin\left(i\pi x^2/2\right) dx \tag{11}$$

which are implemented in standard analysis software. The diffraction pattern of a single slit can thus be written as

$$I \propto \frac{1}{2}\left| F\left(\frac{b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{b-2x}{\sqrt{2\lambda z}}\right) \right|^2 \tag{12}$$

and that of a double slit as

$$I \propto \left| F\left(\frac{g+b+2x}{\sqrt{2\lambda z}}\right) + F\left(\frac{g+b-2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b+2x}{\sqrt{2\lambda z}}\right) - F\left(\frac{g-b-2x}{\sqrt{2\lambda z}}\right) \right|^2 . \tag{13}$$

Often the Fresnel number

$$N_b = \frac{b^2}{4\lambda z} \tag{14}$$

is introduced to characterize the transition between Fresnel and Fraunhofer diffraction. With the Fresnel number the difference of the Fresnel integrals in Eq. (12) can be written as

$$F\left(\sqrt{2}\left(\frac{x}{\sqrt{\lambda z}} + \sqrt{N_b}\right)\right) - F\left(\sqrt{2}\left(\frac{x}{\sqrt{\lambda z}} - \sqrt{N_b}\right)\right) \to 2\sqrt{2N_b}\,\mathrm{sinc}\left(2\pi\sqrt{N_b}\frac{x}{\sqrt{\lambda z}}\right) F'\left(\sqrt{\frac{2}{\lambda z}}x\right) , \tag{15}$$

where the asymptotic form on the right-hand side is valid in the limit $N_b \ll 1$. The unitless Fresnel number can be used in differentiating between the Fraunhofer limit with $N_b \ll 1$ and the Fresnel limit for $N_b > 1$. In case of the double slit, a further Fresnel number

$$N_g = \frac{g^2}{4\lambda z} \tag{16}$$

can be introduced with $N_g > N_b$. In the limit $N_g \ll 1$ the difference of the Fresnel integrals in Eq. (13) is then given by:

$$\begin{aligned} &F\left(\sqrt{2}\left(\frac{x}{\sqrt{\lambda z}}+\sqrt{N_g}+\sqrt{N_b}\right)\right)-F\left(\sqrt{2}\left(\frac{x}{\sqrt{\lambda z}}+\sqrt{N_g}-\sqrt{N_b}\right)\right) \\ &+F\left(\sqrt{2}\left(\frac{x}{\sqrt{\lambda z}}-\sqrt{N_g}+\sqrt{N_b}\right)\right)-F\left(\sqrt{2}\left(\frac{x}{\sqrt{\lambda z}}-\sqrt{N_g}-\sqrt{N_b}\right)\right). \\ &\to 4\sqrt{2N_b}\,\mathrm{sinc}\left(2\pi\sqrt{N_b}\frac{x}{\sqrt{\lambda z}}\right)\cos\left(2\pi\sqrt{N_g}\frac{x}{\sqrt{\lambda z}}\right)F'\left(\sqrt{\frac{2}{\lambda z}}x\right) \end{aligned} \tag{17}$$

In the Fresnel approximation, diffraction at a grating leads to the phenomenon of self-imaging: at regular intervals t_B, where the diffraction pattern looks exactly like the diffraction grating. This is easy to see mathematically. Consider a diffraction grating that is periodic in the x' direction with a period length g, such that the transmission function can be written as a Fourier series:

$$f(x') = \sum_{n=-\infty}^{\infty} c_n \exp\left(-\frac{2\pi i n x}{g}\right). \tag{18}$$

Inserting Eq. (18) into Eq. (10) and integrating over x' yields

$$\psi(x,t) = \exp(ikz - i\omega t)\sum_{n=-\infty}^{\infty} c_n \exp\left(-\frac{2\pi i n x}{g}\right)\exp\left(-i\frac{\pi z \lambda}{g^2}n^2\right). \tag{19}$$

An additional phase factor appears in the sum, which depends on the distance z. If z takes the values

$$z_m = m\frac{2g^2}{\lambda}, \; m \in \mathbb{Z}, \tag{20}$$

the phase factor is equal to one, i.e., the magnitudes of equations (18) and (19) are equal for these distances, and the images look like the diffraction grating.

This self-imaging is also known as the Talbot effect. If the camera is shifted by the Talbot length

$$t_B = \frac{2g^2}{\lambda}, \tag{21}$$

the same image is obtained. Experimentally, however, it is difficult to recognize the similarity of the images. It is therefore more convenient to examine the diffraction patterns in wave vector space and observe the periodicity of the diffraction fringes there.

So far, we have considered a general lattice for which the only assumption was that it is periodic in the x'-direction. In the experiment, line lattices with a spacing of g and a line width of $b < g$ are used. The transmission function of these lattices can be expressed as

$f(x') = 1$ for $x' \in [ng - b/2, ng + b/2]$ und $f(x') = 0$ elsewise, $n \in \mathbb{Z}$.

Inserting in Eq. (10) and integrating yields

$$\psi(x) = \frac{1}{\sqrt{2i}} \exp(ikz - i\omega t) \sum_{n=-\infty}^{\infty} \left[F\left(\frac{2x + 2ng + b}{\sqrt{2\lambda z}} \right) - F\left(\frac{2x + 2ng - b}{\sqrt{2\lambda z}} \right) \right] \tag{22}$$

with the Fresnel integral defined in Eq. (11). The measured intensity is again given by $\psi^* \psi$. The Fourier transform (FT) of the measured intensity can be evaluated using the convolution theorem (see below). For $0 < b \leq g/2$, one obtains:

$$\lambda z |FT| = \begin{cases} \frac{1}{\pi\kappa} \left| \sin\left(\pi\kappa \left(b - \kappa\lambda z \right) \right) \right| |S| & 0 \leq \kappa\lambda z \leq b \\ 0 & b \leq \kappa\lambda z \leq g - b \\ \frac{1}{\pi\kappa} \left| \sin\left(\pi\kappa \left(b - g + \kappa\lambda z \right) \right) \right| |S| & g - b \leq \kappa\lambda z \leq g \end{cases} \tag{23}$$

with the wave number κ. The Fourier transform is periodic in $\kappa\lambda z$ with period g. The structure factor S is given for infinitely large gratings by

$$S = \sum_{n=-\infty}^{\infty} \exp(2\pi i n \kappa g) = \sum_{n=-\infty}^{\infty} \delta(\kappa g - n) \tag{24}$$

i.e. in the Fourier transform sharp peaks appear at $\kappa = n/g, \quad n \in \mathbb{Z}$. Since the grating is finite and since only N slits are illuminated, the structure factor can be approximated by

$$|S| = \left| \sum_{n=-\infty}^{\infty} \exp(2\pi i n \kappa g) \right| = \left| \frac{\sin(N\pi\kappa g)}{\sin(\pi\kappa g)} \right|. \tag{25}$$

Equation (23) takes the form of a structural factor S, which is multiplied by a shape factor. The structural factor S determines that diffraction peaks occur at $\kappa = n/g, \quad n \in \mathbb{Z}$. The amplitude of these diffraction peaks is given by the shape factor. In this case, the shape factor is a $|\sin|$-function that oscillates with the distance z between the grating and the camera. Since the $|\sin|$-function is π-periodic, the function $|FT|$ oscillates with the period

$$\Delta z = \frac{1}{\kappa^2 \lambda} = \frac{g^2}{\lambda n^2} = \frac{t_B}{2n^2}, \tag{26}$$

i.e., the amplitude of the first diffraction peak ($n = 1$) oscillates at half the Talbot length. This provides a way to determine the Talbot length based on the variation in the amplitude of the first diffraction peak.

The formalism introduced here for a two-dimensional lattice reappears in solid-state physics in the context of X-ray diffraction from crystals (Laue scattering), though in a three-dimensional form.

Some properties of the Fourier transform

The Fourier transform of a function $f(x)$ is defined by

$$F(k) = FT\left[f(x)\right] = \int_{-\infty}^{\infty} f(x)\exp(-ikx)dx \tag{27}$$

with the backwards transformation

$$f(x) = \frac{1}{2\pi} FT^{*}\left[F(k)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(k)\exp(ikx)dk\ . \tag{28}$$

In calculations, the representation of the δ-function by a Fourier transformation is often valuable:

$$\delta(x - x_0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \exp(ik(x - x_0))dk\ . \tag{29}$$

If x denotes a spatial coordinate, k has the unit m^{-1}, i.e. k is a wave vector. Therefore, one often refers to real space and wave-vector space (k-space). Accordingly, the Fourier transformation transforms functions from one into the other space.

The definition in equation (14) follows the standard convention commonly used in physics, in which position coordinate and wave vector are conjugate variables. However, there are representations of the Fourier transform using the wave number $\kappa = k/(2\pi)$ *as conjugated variable of the position. With the wave number the definition of the Fourier transform reads*

$$F(\kappa) = FT\left[f(x)\right] = \int_{-\infty}^{\infty} f(x)\exp(-2\pi i\kappa x)dx \tag{27'}$$

In this instruction sheet the physics convention with wave vector will be used.

Let $f(x)$ and $g(x)$ denote functions in real space, $F(k)$ and $G(k)$ the corresponding Fourier transforms in k-space. Then the following relations hold (a, b, c being complex numbers):

1. Parseval's theorem

$$\int_{-\infty}^{\infty} |f(x)|^2\, dx = \int_{-\infty}^{\infty} |F(k)|^2\, dk \tag{30}$$

2. Linearity

$$FT\left[af(x) + bg(x)\right] = aF(k) + bG(k) \tag{31}$$

3. Scaling law

$$FT\left[f(cx)\right] = \frac{1}{c}F\left(\frac{k}{c}\right) \tag{32}$$

4. Displacement law

$$FT\left[f(x-x_0)\right]=\exp(ikx_0)F(k) \tag{33}$$

5. Convolution theorem

$$FT\left[f(x)\otimes g(x)\right]=FT\left[\int_{-\infty}^{\infty}f(x')g(x-x')dx'\right]=FT\left[f(x)\right]FT\left[g(x)\right]=F(k)G(k) \tag{34}$$

The latter can be inverted to yield

$$FT^{*}\left[F(k)G(k)\right]=f(x)\otimes g(x)=\int_{-\infty}^{\infty}f(x')g(x-x')dx'\,. \tag{35}$$

Since the intensity measured by the camera is proportional to the square of the electric field, the last equation especially means that the Fourier transform of the intensity pattern is proportional to the slit/double slit transmission function convoluted with itself.

Hints to task 1

The diffraction pattern is measured with a CCD line camera, the data are transferred to a computer and are subsequently processed. The first column of the imported data is just the pixel index (1 to 3000 or 1 to 2048, depending on camera), the second column contains values proportional to the intensity. Treat this as arbitrary units. Take care that the laser intensity is sufficiently attenuated by the polarizers to avoid an overload of the camera. As a rule of thumb, the largest measured intensity should not exceed 80% of the maximum possible intensity. The pixel index is converted to a position coordinate by multiplying with the pixel width (7 µm for the "old" and 14 µm for the "new" cameras, see additional material). The pixels are at most 200 µm high, i.e. in the Fraunhofer mode the heights of the camera and of the collimated laser beam have to be well adjusted.

Slides with several single slits (usually denoted by A, B, C) are available in the lab. Select one of these slits for your measurements and write down your choice in the lab report. The Fraunhofer diffraction pattern can be observed with the CCD camera placed in the focal plane of a collimating lens, see Fig. 2. Carefully adjust the optical elements in height and orientation to obtain a good intensity profile.

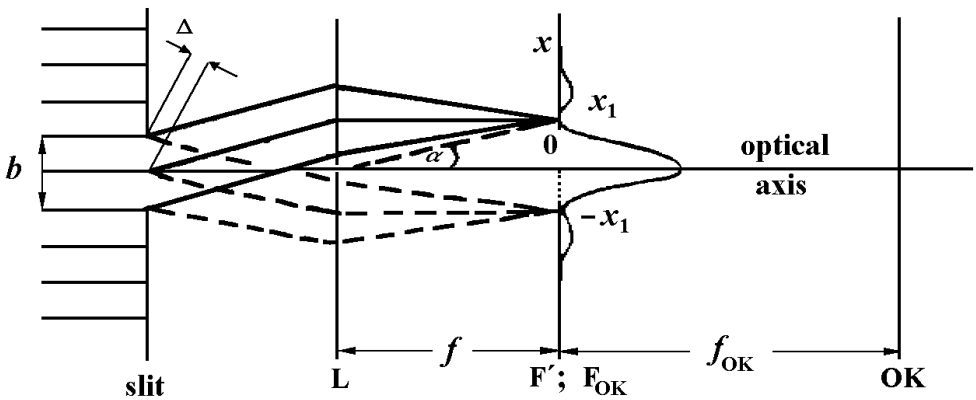

Fig. 2. Fraunhofer diffraction. **L**: lens with focal length *f*; the CCD camera is placed into the focal plane of the lens, indicated by the diffraction pattern. Optionally, the diffraction pattern can be viewed directly through an eyepiece (OK: "Okular"), when the camera is removed. However, since the risk to expose the eye directly to the laser beam is too high, the use of this method is discouraged.

In the observation plane of the CCD camera a diffraction pattern is generated by interference of diffracted waves. The diffraction-pattern minima of arbitrary order n (n = ±1, ± 2, ...) obey the relation

$$b \sin\alpha_n = n\lambda \,, \tag{36}$$

where α_n are the deflection angles under which the diffraction minima appear.

The minima are visible in the focal plane **F'** at positions

$$x_n = f \tan\alpha_n = f \tan\left[\arcsin\left(\frac{n\lambda}{b}\right)\right]. \tag{37}$$

In the small angle approximation $x_n \approx f\frac{n\lambda}{b}$ is valid.

In 1(a) determine the distances $2x_n$ between the minima of $\pm nth$ order and from these obtain the slit width by linear regression.

In 1(b) calculate the FFT of the intensity profile. First shift the intensity distribution along the abscissa such that the principal maximum is located at $x = 0$. The intensity profile is recorded in real space, but can be easily converted into an intensity profile in wave vector space by relating the deflection x of the photons to the wave vector k_x perpendicular to the optical axis by

$$k_x = \frac{2\pi}{\lambda}\sin\alpha = \frac{2\pi}{\lambda}\frac{x}{\sqrt{x^2+f^2}} \approx \frac{2\pi}{\lambda}\frac{x}{f}. \tag{38}$$

Plot the intensity as a function of k_x, perform the FFT and determine b.

In 1(c), the Fresnel diffraction regime is realized by removing the lens and placing the camera directly at a distance z behind the slit. Record diffraction patterns at three different distances and compare with the theory, Eq. (12). Use the slit width b determined from the measurement in the Fraunhofer regime.

You might also try to reach the Fraunhofer regime without the use of a lens, just by placing the camera sufficiently far from the slit. It is, however, not clear, whether the optical bench is long enough to securely do this for all slit widths.

Hints to task 2

The double slit interference pattern given in Eq. (8) consists of a $\cos^2$-term from the interference of the waves emitted by the two slits, modulated by the sinc^2-term from the single-slit diffraction.

When the path difference between the light waves emitted by the two slits is equal to $(n+1/2)\lambda$ with $n = 0, \pm1, \pm2, \ldots$, the waves interfere destructively and the $\cos^2$-term vanishes for the following angles:

$$g\sin\alpha_n = \left(n + \tfrac{1}{2}\right)\lambda \ . \tag{39}$$

The corresponding positions x_n on the camera are obtained from Eq. (23).

The modulation of the double slit interference pattern by the single slit diffraction pattern is illustrated in Fig. 3. The minima of the corresponding envelope function (blue curve in Fig.3) obey the same relation as for a single slit, and might be used to estimate the slit width. However, it is more accurate to fit Eq. (8) to the data to determine both slit distance g and slit width b.

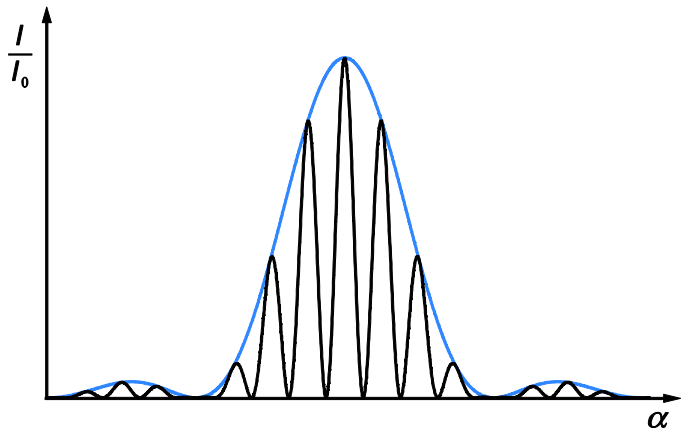

Fig. 3. Diffraction pattern of a double slit (black: double slit interference pattern, blue: envelope function from single slit diffraction).

In 2(a) determine the distances $2x_n$ between corresponding minima and from these obtain the slit distance g by linear regression.

In 2(b) fit Eq. (8) to the data and determine g and b

In 2(c) calculate the FFT of the intensity profile in the same way as explained for the single slit. Determine g and b.

In 2(d), the Fresnel diffraction regime is realized by removing the lens and placing the camera directly at a distance z behind the double slit. Record diffraction patterns at three different distances and compare with the theory, Eq. (13). Use the slit distance g and width b determined from the measurement in the Fraunhofer regime.

Hints to task 3

In task 3, Fresnel diffraction is achieved by removing the lens and placing the camera directly behind the grating at a distance z. Take diffraction images for different distances. Calculate the Fourier transforms of the diffraction images and plot them as a function of the wave number. Determine the grating constant g from the positions of the maxima and compare the experimental value with the manufacturer's specification. Determine the amplitude of the first FT peak and plot it as a function of the distance z. Determine the Talbot length and compare the experimental and theoretical values.

Further information

Manual Thorlabs CCD camera: https://home.uni-leipzig.de/prakphys/download/thorlabsCCD-eng.pdf